

Ex. 7.1

a)

~~$ax^2 + bx + c$~~

$ax^2 + bx + c = y$

$p(x) = x^2$

$$\begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix} \begin{bmatrix} c \\ b \\ a \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} c \\ b \\ a \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

~~$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{array} \right]$~~

~~$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & -1 & -1 \\ 0 & 2 & 0 & 0 \end{array} \right]$~~

$c=0$
 $b=0$
 $a=1$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & -1 & -1 \\ 0 & 2 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

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Ex. 2.1.

$$b) \quad p_{n-1}(x) = y_1 l_1(x) + y_2 l_2(x) + \dots + y_n l_n(x)$$

$$l_j(x) = \frac{\prod_{k=1, k \neq j}^n (x - x_k)}{\prod_{k=1, k \neq j}^n (x_j - x_k)}$$

$$n=3$$

$$l_1(x) = \frac{\cancel{x+1} (x-x_2)(x-x_3)}{(x_1-x_2)(x_1-x_3)} = \frac{\frac{x \cdot (x-1)}{2} \cdot \cancel{(x+1)(x-1)}}{(-1) \cdot \cancel{(-2)}} = \frac{(x+1)(x-1)}{2}$$

$$l_2(x) = \frac{(x-x_1)(x-x_3)}{(x_2-x_1)(x_2-x_3)} = \frac{(x+1)(x-1)}{(1) \cdot (-1)} = -(x+1)(x-1)$$

$$l_3(x) = \frac{(x-x_1)(x-x_2)}{(x_3-x_1)(x_3-x_2)} = \frac{(x+1)x}{2 \cdot 1} = \frac{x(x+1)}{2}$$

$$x_1 = -1$$

$$x_2 = 0$$

$$x_3 = 1$$

$$p_{n-1}(x) = \frac{x \cdot (x-1)}{2} + \frac{x(x+1)}{2} = x^2$$

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Ex. 7.1.

$$c) \pi_j(t) = \prod_{k=1}^{j-1} (t - t_k), \quad j=1, \dots, n.$$

$$p_{n-1}(x) = x_1 \pi_1(x) +$$

$$c) \pi_j(x) = \prod_{k=1}^{j-1} (x - x_k) \quad p_{n-1}(x) = \frac{t_1}{t_1} + \frac{t_2}{t_2} \pi_2(x) + \dots + \frac{t_n}{t_n} \pi_n(x)$$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & (x_2 - x_1) & 0 \\ 1 & (x_3 - x_1) & (x_3 - x_2) \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & (x_2 - x_1) & 0 \\ 1 & (x_3 - x_1) & (x_3 - x_2) \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$t_1 = y_1 = 1$$

$$t_2 = y_2 - y_1 \cdot t_1$$

$$p_1(x) = 1, \quad p_2(x) = 1 + (-1)$$

$$t_2 = \frac{y_2 - t_1}{(x_2 - x_1)} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1}{1} = -1$$

$$t_3 = \frac{y_3 - t_1 - t_2 \cdot (x_3 - x_1)}{(x_3 - x_1)(x_3 - x_2)} = \frac{1 - 1 + 2}{2 \cdot 1} = 1$$

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$$\begin{aligned}
 c) \quad p_2(x) &= 1 + (-1) \cdot (x - x_1) + 1 \cdot (x - x_1)(x - x_2) = \\
 &= 1 - (x + 1) + (x + 1)x = \\
 &= x^2 + (x + 1) - (x + 1) = x^2
 \end{aligned}$$

Ex. 2.5

a)

$$p(x) = t^3 + 3t^2 + 2t + 5$$

$$\begin{bmatrix} 1 & t_1 & t_1^2 & t_1^3 \\ 1 & t_2 & t_2^2 & t_2^3 \\ 1 & t_3 & t_3^2 & t_3^3 \\ 1 & t_4 & t_4^2 & t_4^3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & | & 11 \\ 1 & 2 & 4 & 8 & | & 29 \\ 1 & 3 & 9 & 27 & | & 65 \\ 1 & 4 & 16 & 64 & | & 125 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & | & 5 \\ 0 & 1 & 0 & 0 & | & 2 \\ 0 & 0 & 1 & 0 & | & 3 \\ 0 & 0 & 0 & 1 & | & 1 \end{bmatrix}$$

↑

$$\begin{bmatrix} 1 & 1 & 0 & 0 & | & 7 \\ 0 & 1 & 0 & 0 & | & 2 \\ 0 & 0 & 1 & 0 & | & 3 \\ 0 & 0 & 0 & 1 & | & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & | & 11 \\ 0 & 1 & 3 & 7 & | & 18 \\ 0 & 2 & 8 & 26 & | & 59 \\ 0 & 3 & 15 & 63 & | & 114 \end{bmatrix}$$

$$114 - 59 = 100 - 40 \quad \uparrow$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & | & 11 \\ 0 & 1 & 3 & 7 & | & 18 \\ 0 & 0 & 2 & 12 & | & 18 \\ 0 & 0 & 6 & 42 & | & 60 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & | & 11 \\ 0 & 1 & 3 & 7 & | & 18 \\ 0 & 0 & 1 & 6 & | & 9 \\ 0 & 0 & 0 & 1 & | & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 0 & | & 10 \\ 0 & 1 & 3 & 0 & | & 11 \\ 0 & 0 & 1 & 0 & | & 3 \\ 0 & 0 & 0 & 1 & | & 1 \end{bmatrix}$$

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6)

$$l_1(t) = \frac{(t-t_2)(t-t_3)(t-t_4)}{(t_1-t_2)(t_1-t_3)(t_1-t_4)} = \frac{(t-2)(t-3)(t-4)}{(-1)(-2)(-3)} = \frac{(t-2)(t-3)(t-4)}{-6}$$

$$l_2(t) = \frac{(t-t_1)(t-t_3)(t-t_4)}{(t_2-t_1)(t_2-t_3)(t_2-t_4)} = \frac{(t-1)(t-3)(t-4)}{(-1)(-2)} = \frac{(t-1)(t-3)(t-4)}{2}$$

$$l_3(t) = \frac{(t-t_1)(t-t_2)(t-t_4)}{(t_3-t_1)(t_3-t_2)(t_3-t_4)} = \frac{(t-1)(t-2)(t-4)}{2 \cdot (-1)} = -\frac{(t-1)(t-2)(t-4)}{2}$$

$$l_4(t) = \frac{(t-t_1)(t-t_2)(t-t_3)}{(t_4-t_1)(t_4-t_2)(t_4-t_3)} = \frac{(t-1)(t-2)(t-3)}{3 \cdot 2} = \frac{(t-1)(t-2)(t-3)}{6}$$

$$l_j(t) = \prod_{k=1, k \neq j}^n \frac{(t-t_k)}{(t_j-t_k)}$$

$$p(t) = 11 \cdot \frac{(t-2)(t-3)(t-4)}{(-6)} + 29 \cdot \frac{(t-1)(t-3)(t-4)}{2} - 65 \cdot \frac{(t-1)(t-2)(t-4)}{2} + 125 \cdot \frac{(t-1)(t-2)(t-3)}{6}$$

$$= 29 \cdot \frac{(t-1)(t-3)(t-4)}{2}$$

$$(t-3)(t-4) \left(\frac{29}{2}(t-1) - \frac{11}{6}(t-2) \right) + \frac{(t-1)(t-2)}{2} \left(\frac{125(t-3) - 195(t-4)}{3} \right) =$$

$$= \frac{(t-3)(t-4)}{2} \left(\frac{87t - 87 - 11t + 22}{3} \right) + \frac{(t-1)(t-2)}{6} \left(\frac{195t - 125 - 195t + 780}{3} \right) =$$

$$= \frac{(t-3)(t-4)}{6} (76t - 65) + \frac{(t-1)(t-2)}{6} (70t - 405)$$

c)

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 11 \\ 1 & 1 & 0 & 0 & 29 \\ 1 & 2 & 2 & 0 & 65 \\ 1 & 3 & 6 & 6 & 125 \end{array} \right]$$

$$x_1 = 11$$

$$x_2 = 18$$

$$x_3 = \frac{65 - 11 - 36}{2} = 9$$

$$x_4 = \frac{125 - 11 - 3 \cdot 18 - 6 \cdot 9}{6} = \frac{114 - 54}{6} = 9 = 1$$

$$p(x) = 11 + 18(t-1) + 9(t-1)(t-1) + (t-1)(t-1)(t-9) =$$

$$= 11 + 18t - 18 \cdot 11 + (t-1)(t-1) \cdot t =$$

$$= t^3 + 18t + 18 \cdot 11t$$

$$p(x) = 11 + 18(t-1) + 9(t-1)(t-2) + (t-1)(t-2)(t-3) =$$

$$= 18t - 9 + 9(t^2 - 3t + 2) + (t-1)(t^2 - 5t + 6) =$$

$$= 18t - 9$$

~~$p(t) = \frac{1}{6}$~~

8)

$$\begin{aligned} (t-3)(t-4)(76t-65) &= \cancel{76} (t^2 - 7t + 12)(76t - 65) = \\ &= 76t^3 - 532t^2 + 912t - 65t^2 + 455t - 780 = \\ &= 76t^3 - 597t^2 + 1367t - 780 \end{aligned}$$

$$\begin{aligned} (t-1)(t-2)(70t-405) &= (t^2 - 3t + 2)(70t - 405) = \\ &= 70t^3 - 210t^2 + 140t - 405t^2 + 1215t - 810 = \\ &= 70t^3 - 615t^2 + 1355t - 810 \end{aligned}$$

$$p(t) = \frac{1}{6} (6t^3 + 18t^2 + 12t + 30) = t^3 + 3t^2 + 2t + 5$$

c)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1(t_2-t_1) & 0 & 0 & 0 \\ 1(t_3-t_1) & (t_3-t_1)(t_3-t_2) & 0 & 0 \\ 1(t_4-t_1) & (t_4-t_1)(t_4-t_2) & (t_4-t_1)(t_4-t_3) & (t_4-t_1)(t_4-t_2)(t_4-t_3) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

c)

$$p(x) = 11 + 10(1-1) + 18 + 9(1-2) + (1-1)(1-2)(1-3) =$$

$$= 11 + (1-1)(9) + (1-1)(1-1)(1-3) =$$

$$= 11 + (1-1)(9 + 1^2 - 5 + 6) =$$

$$= 11 + (1-1)(1^2 + 4 + 6) = 11 + 1^3 + 4 \cdot 1^2 + 6 \cdot 1 - 1^2 - 4 \cdot 1 - 6 =$$

$$= \boxed{1^3 + 3 \cdot 1^2 + 2 \cdot 1 + 5}$$

c) incremental:

$$x_{i+1} = \frac{y_{i+1} - p_i(t_{i+1})}{\pi_{i+1}(t_{i+1})}$$

$$x_1 = 11 = y_1$$

$$x_2 = \frac{y_2 - p_1(t_2)}{(t_2 - t_1)} = \frac{29 - 11}{1} = 18$$

$p_1(t_3) = x_1 + x_2(t_3 - t_1)$
 $* x_3(t_4 - t_1)(t_4 - t_2)$

$$x_3 = \frac{y_3 - p_2(t_3)}{(t_3 - t_1)(t_3 - t_2)} = \frac{55 - 11 - 18 \cdot 2}{2} = 9$$

$$x_4 = \frac{y_4 - p_3(t_4)}{(t_4 - t_1)(t_4 - t_2)(t_4 - t_3)} = \frac{125 - 11 - 18 \cdot 3 - 9 \cdot 5}{6} = 1$$

$p(t) = t^3 + 3t^2 + 2t + 5$

the $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ is same with the solution of triangular matrix, so, $p(t) = t^3 + 3t^2 + 2t + 5$

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Ex. 7:5.

B.

c) divided diff.

$$f[x_1 \dots x_k] = \frac{f[x_2 \dots x_k] - f[x_1 \dots x_{k-1}]}{t_k - t_1}$$

$$f[x_1] = y_1 = 11$$

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{t_2 - t_1} = 18$$

~~$f[x_1, x_2, x_3]$~~

$$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{t_3 - t_2} = 36.$$

$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ is same with other 2 methods, so,

$$p(x) = t^3 + 3t^2 + 2t + 5$$

$$f[x_1, x_2, x_3] = \frac{\cancel{f[x_3]} - f[x_2, x_3] - f[x_1, x_2]}{t_3 - t_1} = \frac{18}{2} = 9$$

$$f[x_2, x_3, x_4] = \frac{f[x_3, x_4] - f[x_2, x_3]}{t_4 - t_2} = \frac{60 - 36}{2} = 12.$$

$$f[x_3, x_4] = \frac{f[x_4] - f[x_3]}{t_4 - t_3} = \frac{60}{1} = 60$$

$$f[x_1, x_2, x_3, x_4] = \frac{f[x_2, x_3, x_4] - f[x_1, x_2, x_3]}{t_4 - t_1} = \frac{12 - 9}{3} = 1$$

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Ex. 2.10.

$$a) \pi(t) = (t - t_j) \prod_{i=1, i \neq j}^n (t - t_i) = (t - t_j) \Gamma(t)$$

$$\Gamma(t) = \prod_{i=1, i \neq j}^n (t - t_i)$$

we need to prove:

$$\pi(t) = \prod_{k=1}^n (t - t_k) \quad (1)$$

$$\pi'(t_j) = \Gamma(t_j). \quad 0 \text{ as } t_j - t_j = 0.$$

$$\pi'(t_j) = (t_j - t_j) \Gamma(t)' \Big|_{t=t_j} + (t - t_j)' \Big|_{t=t_j} \Gamma(t_j) =$$

$$= (t - t_j)' \Big|_{t=t_j} \Gamma(t_j) = \Gamma(t_j) = \prod_{i=1, i \neq j}^n (t_j - t_i) \quad (2)$$

b)

$$l_j(t) = \prod_{k=1, k \neq j}^n \frac{(t - t_k)}{(t_j - t_k)} = \frac{\prod_{k=1, k \neq j}^n (t - t_k)}{\prod_{k=1, k \neq j}^n (t_j - t_k)} =$$

$$= \frac{(t - t_j) \prod_{k=1, k \neq j}^n (t - t_k)}{(t - t_j) \prod_{i=1}^n (t_j - t_i)} = \frac{\prod_{k=1}^n (t - t_k)}{(t - t_j) \pi'(t_j)} = \boxed{\frac{\pi(t)}{(t - t_j) \pi'(t_j)}}$$

using $i=k$, and (1), and (2)

Ex. 7.9.

Sec. 5.5.5.

we use $\begin{bmatrix} f_a \\ f_b \\ f_c \end{bmatrix}$ as $\begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix}$

and $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ as $\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$

we are interested in finding

$p(t)$, of Lagrange where

$p_2(t)$ is Lagrange interpolation

using $\begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix}$ data.

~~$p_2(t) = x_1 + x_2(t-t_1) + x_3(t-t_1)(t-t_2)$~~

$p_2(t) = y_1 l_1(t) + y_2 l_2(t) + y_3 l_3(t) = p(t)$

$l_1(t) = \frac{(t-t_2)(t-t_3)}{(t_1-t_2)(t_1-t_3)}$

$l_2(t) = \frac{(t-t_1)(t-t_3)}{(t_2-t_1)(t_2-t_3)}$

$l_3(t) = \frac{(t-t_1)(t-t_2)}{(t_3-t_1)(t_3-t_2)}$

~~$p_2(t) = y_1 \frac{t_2 t_3}{(t_1-t_2)(t_1-t_3)}$~~

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Ex. 7.9.

$$p(\theta) = y_1 \frac{t_2 t_3}{(t_1 - t_2)(t_1 - t_3)} + y_2 \frac{t_1 t_3}{(t_2 - t_1)(t_2 - t_3)} + y_3 \frac{t_1 t_2}{(t_3 - t_1)(t_3 - t_2)} =$$

$$= y_1 \frac{1}{\left(\frac{t_1}{t_2} - 1\right)\left(\frac{t_1}{t_3} - 1\right)} =$$

$$= y_1 \frac{-t_2}{t_1 \left(1 - \frac{t_2}{t_1}\right) \left(1 - \frac{t_1}{t_3}\right)} + y_2 \frac{1}{\left(1 - \frac{t_2}{t_1}\right) \left(1 - \frac{t_2}{t_3}\right)} + y_3 \frac{t_1 t_2}{t_3^2 \left(1 - \frac{t_1}{t_3}\right) \left(1 - \frac{t_1}{t_3}\right)} =$$

$$= \cancel{a} y_1 \frac{-v}{(1-v)(1-u)} + y_2 \frac{1}{(1-v)(1-u)} + \frac{y_3 \cdot \omega \cdot u}{(1-u)(1-u)} =$$

$$= \frac{-av(1-u) + b(1-u) + c\omega u(1-v)}{(1-v)(1-u)(1-u)} =$$

$$= \frac{av(1-u) + \cancel{b(1-u)} + b(\omega-1) + c\omega u(v-1)}{(v-1)(\omega-1)(u-1)} =$$

$$= b + \frac{av(1-u) + b(\omega-1) - b(\omega-1)(v-1)(u-1) + c\omega u(v-1)}{(v-1)(\omega-1)(u-1)} =$$

$$\text{Ex. 2.9.} \quad v = \frac{u}{\omega} \quad \omega(v-1) = u - \omega$$

$$b + \frac{v(a(1-u) + c\omega(u-\omega) + b(\omega-1)(\frac{1}{b} - (1-\frac{1}{b})(u-1)))}{q} =$$

$$= b + \frac{v(a(1-u) + c\omega(u-\omega) + b(\omega-1)(\frac{\omega}{u} - u + 1 + \frac{u}{b} - \frac{\omega}{u}))}{q} =$$

$$= b + \frac{v(a(1-u) + c\omega(u-\omega) + b(\omega-1)((\omega-u)+1))}{q} =$$

$$= b + \frac{v(a(1-u) + c\omega(u-\omega) + b\omega(\omega-1) - b(\omega-1)(u-1))}{q}$$

$$b + \frac{v(a(1-u) + c\omega(u-\omega) + b\omega(\omega-u) - \cancel{b(\omega-1)(u-1)} + \cancel{b(\omega-1)(u-1)})}{q} =$$

$$= b + \frac{v(a(1-u) + \omega(u-\omega)(c-b) + bu - b)}{q} =$$

$$= b + \frac{v((a-b)(1-u) + \omega(u-\omega)(c-b))}{q} =$$

$$= b + \frac{v(\omega(u-\omega)(c-b) - (1-u)(b-a))}{q} = \boxed{b + \frac{p}{q}}$$